

# Introduction to Bioinformatics

Day 2: Sequence Alignment and  
Phylogenetic  
13<sup>th</sup> January 2026

# Outline Day 2

**Morning (9-12 pm):**

## **Pairwise and Multiple Sequence Alignment:**

- **Overview of alignment concepts:** Pairwise alignment (global vs local) and multiple sequence alignment
- **Algorithm**
  - **Global Alignments:** Needleman-Wunsch algorithm
  - **Local Alignments:** Smith-Waterman algorithm
  - **Multiple Sequence Alignment:** ClustalW and Clustal Omega

**Practical Session:**

## **Primary option:**

- Use NCBI BLAST (web-based) to perform local alignment and sequence similarity searches
- Perform multiple sequence alignment using Clustal Omega (web-based)
- Code with python

**Afternoon (2-5 pm): Phylogenetics**

## **Phylogenetics:**

- Introduction to Phylogenetics tree concept, and significance in evolution biology.

## **Practical Session:**

- Construct phylogenetics trees using Clustal Omega (web-based) starting from multiple sequence alignments.

Code with python

# Day 2: Sequence Alignment and Phylogenetic

## Morning Session

# What is a Sequence?

Specific linear order of nucleotides (in DNA or RNA) or amino acids (in proteins) that determines the structure and function of these biomolecules.

**DNA Sequence:** The arrangement of bases (adenine [A], thymine [T], cytosine [C], guanine [G]).

**RNA Sequence:** The arrangement of bases (adenine [A], uracil [U], cytosine [C], guanine [G]).

**Protein Sequence:** The arrangement of amino acids in a polypeptide chain.

Sequences encode genetic information and are fundamental to biological processes like replication, transcription, and translation.



Sample genetic code with complementary strands.

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# Examples of Sequences?

DNA Sequence: order of nucleotide bases (adenine, thymine, cytosine, guanine)

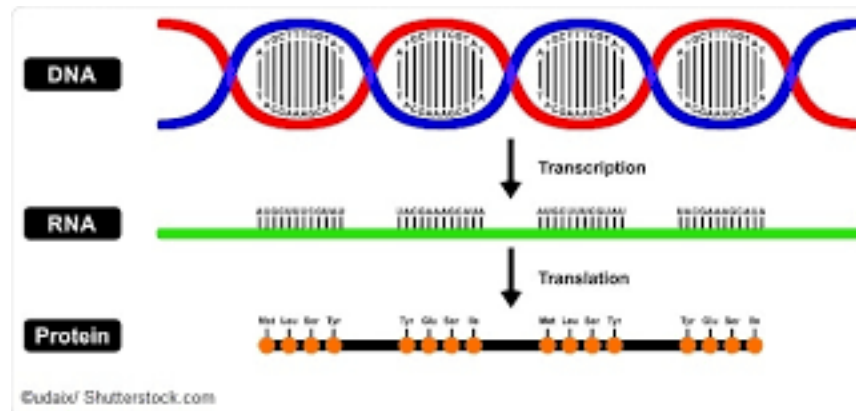
Example. **ATCGTACGGA**

RNA Sequence: Similar to, DNA1 but thymine (T) is replaced by uracil (U) 'in RNA.

Example: **AUCGAUCGGA**

Protein Sequence: order ,of amino acids (e.g., methionine, alanine, leucine,,glycine)

Example· MET-ALA-LEU-GLV



# FASTA

A text-based file format used to represent nucleotide or protein sequences. It is commonly used in bioinformatics to store and exchange sequence data. A FASTA file consists of:

- 1. Header Line:** Starts with a > symbol, followed by an identifier or description of the sequence.
- 2. Sequence Data:** The following lines contain the actual sequence, either nucleotide (DNA/ RNA) or protein, and can span multiple lines.

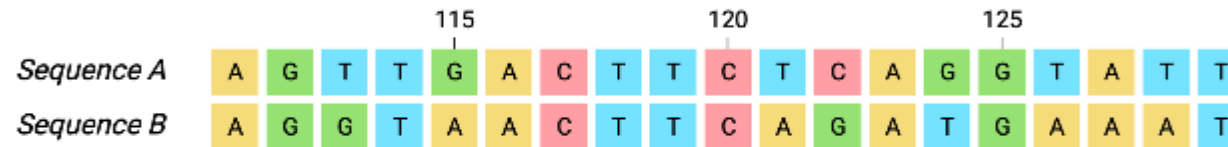
Example:

```
>Sequence_ID_1 Description of the first sequence  
AGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTA
```

```
>Sequence_ID_2 Description of the second sequence  
TGCATGCATGCATGCATGCGTACGATCGTACG
```

# What is Sequence Alignment?

Sequence alignment is the process of arranging two or more biological sequences such as nucleotide (DNA, RNA), or amino acid (protein) sequences to identify regions of similarity.



# What alignments can help?



DETERMINE EVOLUTIONARY  
RELATIONSHIPS AMONG GENES,  
PROTEINS, AND SPECIES



DETERMINE FUNCTION OF A  
NEWLY DISCOVERED GENE  
SEQUENCE



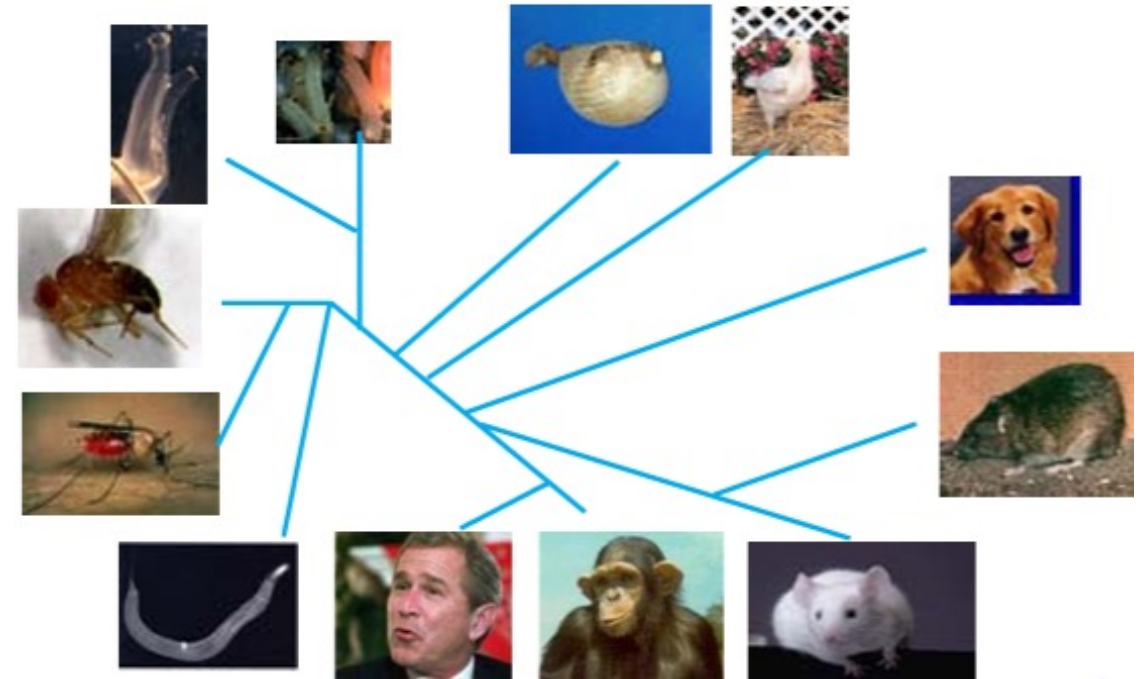
PREDICTING STRUCTURE AND  
FUNCTION OF PROTEIN



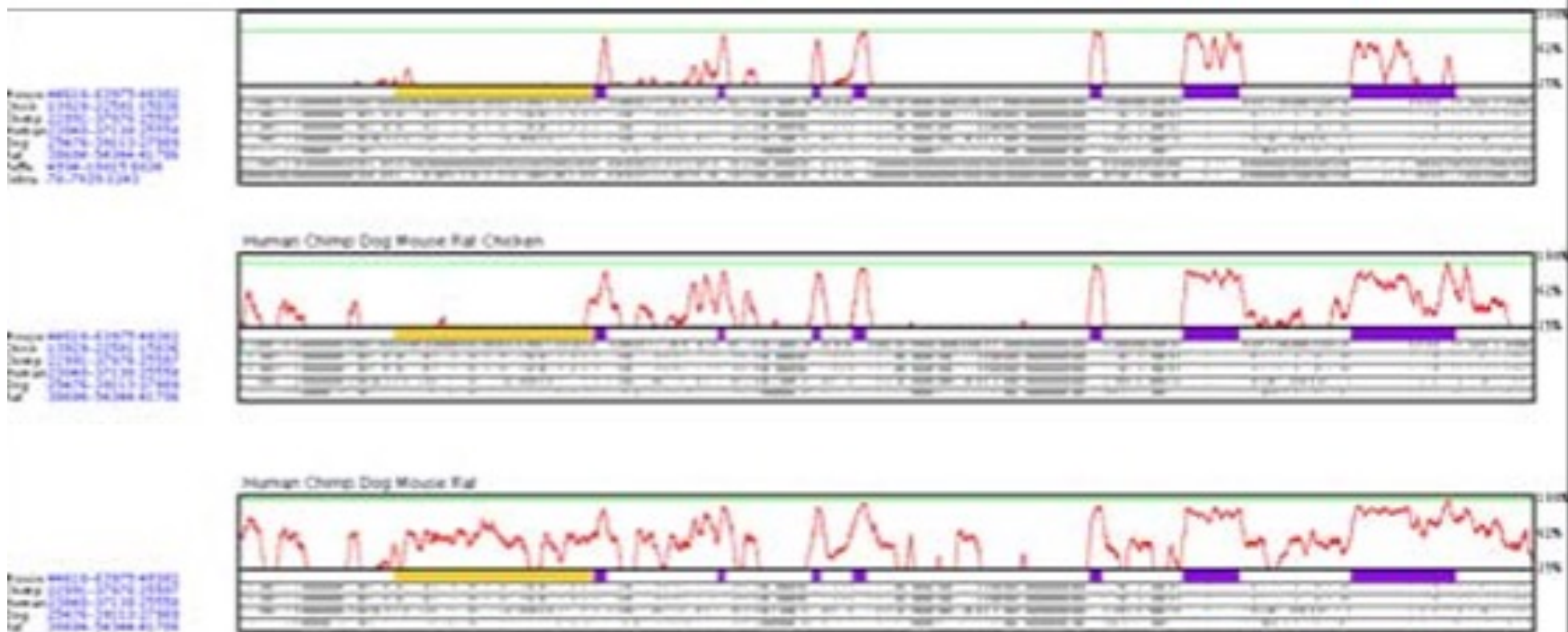
# Importance of Sequence Alignment in Bioinformatics

## Evolutionary Relationships

- Helps in identifying similarities between genes/  
proteins from different species.
- Homologous sequences, (genes from a common ancestor) can be identified through alignment.
- Helps inferring evolutionary distances using substitution patterns (changes in nucleotides).
- Helps in phylogenetic tree construction



# Sequence conversation implies function



**Alignment is the key to**

- Finding important regions
- Determining function
- Uncovering the evolutionary forces



# Algorithm used in Dynamic programming

**Needleman-wunsch Algorithm**

**Developed by Saul.b. needleman  
& christian.d. wunsch**

**Referred as global alignment**

**Used in aligning 2 closely related  
sequence**

**Compares the whole sequence**

**Tools:- EMBOSS-Needle,  
Specialised BLAST**

**Smith-Waterman algorithm**

**Developed by Temple.F.Smith  
& Michael. S. Waterman**

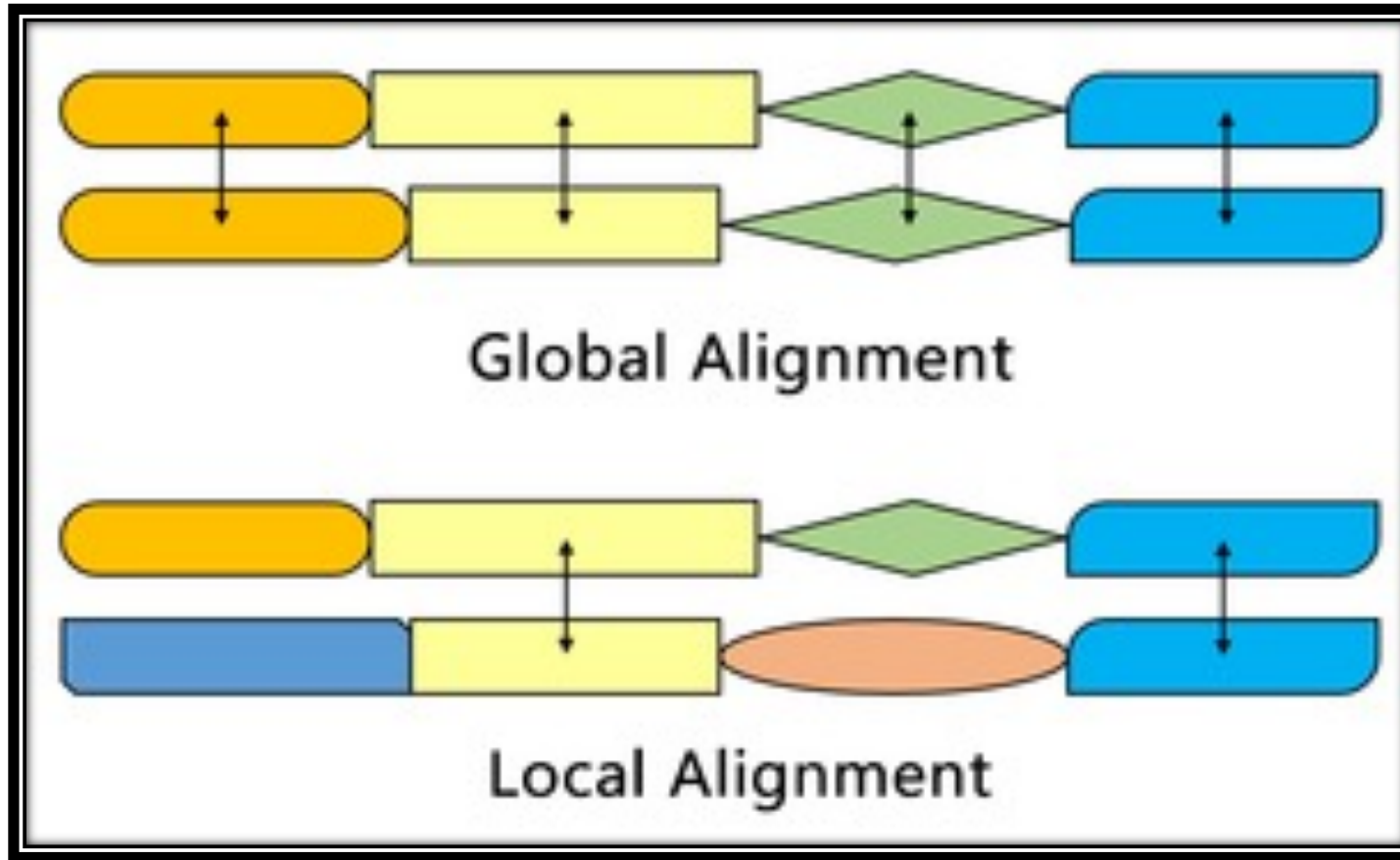
**Referred as local alignment**

**Used in aligning divergent  
sequences**

**Compares a patch from the  
sequence**

**Tools:-EMBOSS-Water,  
LALIGN**

# Comparison of Global and Local alignment in general



# Needleman-Wunsch algorithm

- Aligns protein or nucleic acid sequences
- Divides a large problem into a series of smaller problems
- Uses the solution of smaller problem to reconstruct a solution to the larger problems

# Constructing the matrix

We will have 2 matrices of 2D representation viz,

1. The score matrix
2. Traceback matrix

The N-W algorithm consists of 4 steps:-

1. Initialization of the score matrix
2. Filling up the matrix
3. Traceback
4. Alignment

# 1. Initializing the scoring matrix

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  |    |    |    |    |     |     |
| C | -4  |    |    |    |    |     |     |
| T | -6  |    |    |    |    |     |     |
| A | -8  |    |    |    |    |     |     |
| G | -10 |    |    |    |    |     |     |
| C | -12 |    |    |    |    |     |     |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | -1          |
| Gap Penalty    | -2          |

## 2. Filling the matrix

|   |    |    |
|---|----|----|
|   |    | G  |
|   | 0  | -2 |
| G | -2 | X  |

For x :

This cell has 3 possible values

- Top :-  $(-2)+(-2) = -4$
- Left :-  $(-2)+(-2) = -4$
- Top-left :-  $(0)+(1) = 1$

The highest value is 1 and thus it is entered into the cell  
i.e  $x = 1$

|   |    |          |
|---|----|----------|
|   |    | G        |
|   | 0  | -2       |
| G | -2 | <b>1</b> |



# Contd..

|          |    |          |          |
|----------|----|----------|----------|
|          |    | <b>G</b> | <b>C</b> |
|          | 0  | -2       | -4       |
| <b>G</b> | -2 | 1        | <b>X</b> |
| <b>C</b> | -4 | <b>Y</b> |          |

For **X** :

Top:  $(-4)+(-2) = (-6)$

Left:  $(+1)+(-2) = (-1)$

Top-Left:  $(-2)+(-1) = (-3)$

For **Y** :

Top:  $(1)+(-2) = (-1)$

Left:  $(-4)+(-2) = (-6)$

Top-Left:  $(-2)+(-1) = (-3)$

The highest value for X and Y is  
-1, thus it is entered into the cell.  
i.e **X = -1; Y = -1**

|          |    |           |           |
|----------|----|-----------|-----------|
|          |    | <b>G</b>  | <b>C</b>  |
|          | 0  | -2        | -4        |
| <b>G</b> | -2 | 1         | <b>-1</b> |
| <b>C</b> | -4 | <b>-1</b> |           |

# 1. Initializing the scoring matrix

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  |    |    |    |    |     |     |
| T | -6  |    |    |    |    |     |     |
| A | -8  |    |    |    |    |     |     |
| G | -10 |    |    |    |    |     |     |
| C | -12 |    |    |    |    |     |     |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | -1          |
| Gap Penalty    | -2          |

# 1. Initializing the scoring matrix

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  | -1 | 0  | -2 | -2 | -4  | -6  |
| T | -6  |    |    |    |    |     |     |
| A | -8  |    |    |    |    |     |     |
| G | -10 |    |    |    |    |     |     |
| C | -12 |    |    |    |    |     |     |

| Parameter      | Value       |
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| Sequence 1     | G C T A G C |
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# 1. Initializing the scoring matrix

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  | -1 | 0  | -2 | -2 | -4  | -6  |
| T | -6  | -3 | 0  | -1 | -3 | -3  | -5  |
| A | -8  |    |    |    |    |     |     |
| G | -10 |    |    |    |    |     |     |
| C | -12 |    |    |    |    |     |     |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | -1          |
| Gap Penalty    | -2          |

# 1. Initializing the scoring matrix

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  | -1 | 0  | -2 | -2 | -4  | -6  |
| T | -6  | -3 | 0  | -1 | -3 | -3  | -5  |
| A | -8  | -5 | -2 | 1  | -1 | -3  | -4  |
| G | -10 |    |    |    |    |     |     |
| C | -12 |    |    |    |    |     |     |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | -1          |
| Gap Penalty    | -2          |

# 1. Initializing the scoring matrix

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  | -1 | 0  | -2 | -2 | -4  | -6  |
| T | -6  | -3 | 0  | -1 | -3 | -3  | -5  |
| A | -8  | -5 | -2 | 1  | -1 | -3  | -4  |
| G | -10 | -7 | -4 | -1 | 0  | 0   | -2  |
| C | -12 |    |    |    |    |     |     |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | -1          |
| Gap Penalty    | -2          |

# 1. Initializing the scoring matrix

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  | -1 | 0  | -2 | -2 | -4  | -6  |
| T | -6  | -3 | 0  | -1 | -3 | -3  | -5  |
| A | -8  | -5 | -2 | 1  | -1 | -3  | -4  |
| G | -10 | -7 | -4 | -1 | 0  | 0   | -2  |
| C | -12 | -9 | -6 | -3 | 0  | -1  | 1   |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | -1          |
| Gap Penalty    | -2          |

# 3. Traceback

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  | -1 | 0  | -2 | -2 | -4  | -6  |
| T | -6  | -3 | 0  | -1 | -3 | -3  | -5  |
| A | -8  | -5 | -2 | 1  | -1 | -3  | -4  |
| G | -10 | -7 | -4 | -1 | 0  | 0   | -2  |
| C | -12 | -9 | -6 | -3 | 0  | -1  | 1   |

| Step | Position | Seq1 | Seq2 | Action   | Score Contribution |
|------|----------|------|------|----------|--------------------|
| 1    | (6,6)    | C    | C    | Match    | +1                 |
| 2    | (5,5)    | G    | G    | Match    | +1                 |
| 3    | (4,4)    | A    | C    | Mismatch | -1                 |
| 4    | (4,3)    | —    | A    | Gap      | -2                 |
| 5    | (3,3)    | T    | T    | Match    | +1                 |
| 6    | (2,2)    | C    | —    | Gap      | -2                 |
| 7    | (1,1)    | G    | G    | Match    | +1                 |

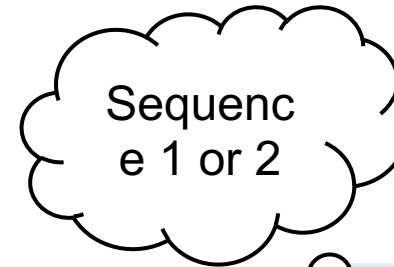


# 4. Alignment

Rules :-

1. If arrow is vertical/horizontal assign a gap and a character

↑ / → :- Gaps and characters  
Where to Assign  
a gap and a character ?



# Contd...

Ans) The gap will be assigned in the direction of the arrow and the character will be assigned in the opposite direction

2. If there is a diagonal arrow both the characters will be assigned

:- Both characters



# Result of alignment

|   | -   | G  | T  | A  | C  | G   | C   |
|---|-----|----|----|----|----|-----|-----|
| - | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| G | -2  | 1  | -1 | -3 | -5 | -7  | -9  |
| C | -4  | -1 | 0  | -2 | -2 | -4  | -6  |
| T | -6  | -3 | 0  | -1 | -3 | -3  | -5  |
| A | -8  | -5 | -2 | 1  | -1 | -3  | -4  |
| G | -10 | -7 | -4 | -1 | 0  | 0   | -2  |
| C | -12 | -9 | -6 | -3 | 0  | -1  | 1   |

| Metric               | Value       |
|----------------------|-------------|
| Optimal Score        | 1           |
| Number of Matches    | 4           |
| Number of Mismatches | 1           |
| Number of Gaps       | 1           |
| Aligned Seq1         | G C T A G C |
| Aligned Seq2         | G - T C G C |

# Smith-Waterman algorithm

- S-W algorithm is modified version of Needleman-Wunsch algorithm
- Negative scoring matrix cells are set to zero
- Traceback procedure starts at the highest scoring matrix cells and procedure until a cell with score zero is found

# Constructing the matrix

|   | – | G | T | A | C | G | C |
|---|---|---|---|---|---|---|---|
| – | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| G | 0 | . | . | . | . | . | . |
| C | 0 | . | . | . | . | . | . |
| T | 0 | . | . | . | . | . | . |
| A | 0 | . | . | . | . | . | . |
| G | 0 | . | . | . | . | . | . |
| C | 0 | . | . | . | . | . | . |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | –1          |
| Gap Penalty    | –2          |

# Contd..

|          |   |          |          |
|----------|---|----------|----------|
|          |   | <b>G</b> | <b>C</b> |
|          | 0 | 0        | 0        |
| <b>G</b> | 0 | 1        | <b>X</b> |
| <b>C</b> | 0 | <b>Y</b> |          |

For X :

Top:  $(0)+(-1) = (-3)$

Left:  $(0)+(-1) = (0)$

Top-Left:  $(-1)+(-1) = (-2)$

For Y :

Top:  $(1)+(-1) = (0)$

Left:  $(-2)+(-1) = (-3)$

Top-Left:  $(-1)+(-1) = (-2)$

The highest value for X and Y is 0,  
thus it is entered into the cell.

i.e  $X = 0$ ;  $Y = 0$

|          |   |          |          |
|----------|---|----------|----------|
|          |   | <b>G</b> | <b>C</b> |
|          | 0 | 0        | 0        |
| <b>G</b> | 0 | 1        | <b>0</b> |
| <b>C</b> | 0 | <b>0</b> |          |

# Constructing the matrix

|   | – | G | T | A | C | G | C |
|---|---|---|---|---|---|---|---|
| – | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| G | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| C | 0 | . | . | . | . | . | . |
| T | 0 | . | . | . | . | . | . |
| A | 0 | . | . | . | . | . | . |
| G | 0 | . | . | . | . | . | . |
| C | 0 | . | . | . | . | . | . |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | –1          |
| Gap Penalty    | –2          |

# Constructing the matrix

|   | - | G | T | A | C | G | C |
|---|---|---|---|---|---|---|---|
| - | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| G | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| C | 0 | 0 | 0 | 0 | 1 | 0 | 2 |
| T | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| A | 0 | 0 | 0 | 2 | 0 | 0 | 0 |
| G | 0 | 1 | 0 | 0 | 1 | 1 | 0 |
| C | 0 | 0 | 0 | 0 | 1 | 0 | 2 |

| Parameter      | Value       |
|----------------|-------------|
| Sequence 1     | G C T A G C |
| Sequence 2     | G T A C G C |
| Match Score    | +1          |
| Mismatch Score | -1          |
| Gap Penalty    | -2          |

Global maximum encountered: 2 (multiple cells: (2,6), (4,3), (6,6)).



# Traceback

|   | - | G | T | A | C | G | C |
|---|---|---|---|---|---|---|---|
| - | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| G | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| C | 0 | 0 | 0 | 0 | 1 | 0 | 2 |
| T | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| A | 0 | 0 | 0 | 2 | 0 | 0 | 0 |
| G | 0 | 1 | 0 | 0 | 1 | 1 | 0 |
| C | 0 | 0 | 0 | 0 | 1 | 0 | 2 |

Alignment

Seq1: G C T A [G C]

Seq2: G T A C [G C]

Alignment

Seq1: [G C] T A G C

Seq2: G T A C [G C]

Alignment

Seq1: G C [T A] G C

Seq2: G [T A] C G C

# Alignment

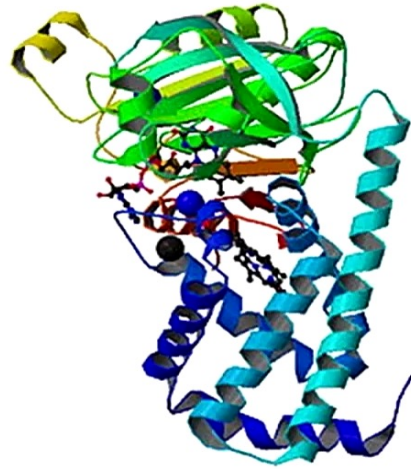
Rules:

Same as of Needleman- Wunsch algorithm

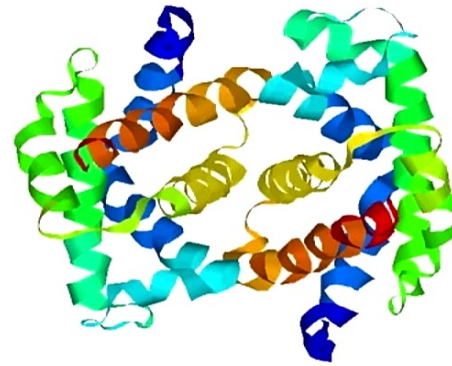
# Difference between the procedure of S-W and N-W algorithm

|                | Smith–<br>Waterman<br>algorithm                         | Needleman–<br>Wunsch<br>algorithm                                          |
|----------------|---------------------------------------------------------|----------------------------------------------------------------------------|
| Initialization | First row and first column are set to 0                 | First row and first column are subject to gap penalty                      |
| Scoring        | Negative score is set to 0                              | Score can be negative                                                      |
| Traceback      | Begin with the highest score, end when 0 is encountered | Begin with the cell at the lower right of the matrix, end at top left cell |

# Example



Flavohemoprotein of  
*Escherichia coli*



haemoglobin subunit alpha  
*Homo sapiens*



MNPYIYLGGAILAEVIGTTLMKFSEGF  
TRLWPSVGTIICYCASFWLLAQTLAYIP  
TGIAYAIWSGVGIVLISLLSWGFFGQRL  
DLPAIIGMMLICAGVLIINLLSRSTPH



MSEALKILNNIRTLRAQARECTLETLEEMLE  
KLEVVVNERREEESAAAAEVEERTRKLQY  
REMLIADGIDPNELLNSLA AVKSGTKAKRA  
QRPAYSYVDENGETKTWTGQGRTPAVIK



To align two sequences of 300 aa/nt lengths,  
approximately  $10^{88}$  comparisons are to be made

# EMBOSS Programs

Feedback

Share

Tools > EMBOSS Programs

## Selected EMBOSS tools for sequence analysis

### Pairwise Sequence Alignment

#### Needle ?

Create an optimal global alignment of two sequences using the Needleman-Wunsch algorithm

Protein Nucleotide

#### Stretcher ?

Improved version of the Needleman-Wunsch algorithm that allows larger sequences to be globally aligned

Protein Nucleotide

#### Water ?

Use the Smith-Waterman algorithm to calculate the local alignment of two sequences

Protein Nucleotide

#### Matcher ?

Emboss homepage

Protein alignment
Nucleotide alignment
Web services
Help & Documentation
Feedback
Share

This is the form for protein sequences. Please go to the [nucleotide form](#) if you wish to align DNA or RNA sequences.

STEP 1 - Enter your protein sequences

Enter or paste your first **protein** sequence in any supported format:

```
>Flavohemoprotein_Escherichia_coli
MLDAQTIATVKATIPLLVETGPKLTAHFYDRMFTHNPELKEIFNMSNQRNGDQREALFNA
IAAYASNIENTPALLPAVEKIAQKHTSFQIKPEQYNIVGEHLLATLDEMSPGQEVLDW
GKAYGVLNVFINREAEIYNENASKAGGWEGTRDFRIVAKTPRSALITSFELEPVDGGAV
AEYRPGQYLGVWLKPEGFPHQEIRQYSLTRKPDGKGYRIAVKREEGGQVSNWLNHANVG
DVVKLVAPAGDFFMAVADDTPTVLISAGVGQTPMLAMLDLAKAGHTAQVNWFFHAAENG
VHAFADDEVKELGQSLPRFTAHTWYRQPSEADRAKGQFDSEGLMDLSKLEGAFSPTMQFY
LCGPVGFQMFTAQQLVDLGVKQENIHYESCFGPHKVL
```

Or, [upload a file](#):  No file chosen

**AND**

Enter or paste your second **protein** sequence in any supported format:

```
>Hemoglobin_subunit_alpha_Homo_sapiens
MVLSPADKTNVKAAGWGVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKGHG
KKVADALTNAVAHVDDMPNALSALSDLHAHKLRLVDPVNFKLLSHCLLVTLAAHLPAEFTP
AVHASLDKFLASVSTVLTSKYR
```

Or, [upload a file](#):  No file chosen

STEP 2 - Set your pairwise alignment options

Entering the sequence to be compared in  
N-W algorithm



| Protein alignment | Nucleotide alignment                                   | Web services | Help & Documentation | Feedback | Share |
|-------------------|--------------------------------------------------------|--------------|----------------------|----------|-------|
| #                 | #                                                      | #            | #                    | #        | #     |
| Flavohemoprot     | 1 M-LDAQTIATVKATIPLLVETGPKLTAHFYDRMFTHNPELKEIFNMSNQR   | 49           |                      |          |       |
| Hemoglobin_su     | 1 MVLSPADKTIIVKAAWGVGAHAGEYGAEALERMFSPPTTKTYFPFDLS     | 50           |                      |          |       |
| Flavohemoprot     | 50 NGDQR-----EALFNAIAAYASNIEINLPALLPAVEKI-AQKHTSFQ     | 89           |                      |          |       |
| Hemoglobin_su     | 51 HGSAQVKGHGKGVADALTNAVA---HVDDMPNALSALSDLHAHK---LR   | 93           |                      |          |       |
| Flavohemoprot     | 90 IKPEQYNIIVGEHLATLDEMFSQGEVLDANGKAYGVLANVFINREAEIY   | 139          |                      |          |       |
| Hemoglobin_su     | 94 VDPVNFKLLSHCLLVTL-----                              | 110          |                      |          |       |
| Flavohemoprot     | 140 NENASKAGGWEGTRDFRIVAKTPRSALITSFELEPVDGGAVAEYRPGQYL | 189          |                      |          |       |
| Hemoglobin_su     | 111 -----AAHLP-----AEFTPA---                           | 121          |                      |          |       |
| Flavohemoprot     | 190 GVNILKPEGFPHQEIRQYSLTRKPDGKGYRIAVKREEGQVSNMLHNHANV | 239          |                      |          |       |
| Hemoglobin_su     | 122 -----                                              | 121          |                      |          |       |
| Flavohemoprot     | 240 GDVVKLVPAGDFFMAVADDTPTVLISAGVGQTPMLAMLDLAKAGHTAQ   | 289          |                      |          |       |
| Hemoglobin_su     | 122 -----                                              | 121          |                      |          |       |
| Flavohemoprot     | 290 VMNFHAAENGDVHFADEVKELGQSLPRFTAHTWYRQPSADRAKGQFDS   | 339          |                      |          |       |
| Hemoglobin_su     | 122 -----VHA-----SLDKFLA-----S                         | 132          |                      |          |       |
| Flavohemoprot     | 340 EGLMDLSKLEGAFSDPTMQFYLCGPVGMQFTAKQLVDLGVKQENIHVEC  | 389          |                      |          |       |
| Hemoglobin_su     | 133 VSTVLTSKYR-----                                    | 142          |                      |          |       |
| Flavohemoprot     | 390 FGPHKVL                                            | 396          |                      |          |       |
| Hemoglobin_su     | 143 -----                                              | 142          |                      |          |       |
| #                 | #                                                      | #            | #                    | #        | #     |

Protein alignment result



# Pairwise sequence alignment

- When comparing 2 sequences it is **Pairwise sequence alignment** (nucleic acids or protein)
- When comparing more than 2 sequence it is **Multiple sequence alignment** (nucleic acids or protein)

# Contd...

- Pairwise sequence alignment is concerned with comparing 2 DNA or 2 Amino acids sequences
- For ex.
  - BLASTn :- for nucleotide sequence
  - BLASTp :- for protein sequence

# BLAST-n and BLAST-P

- BLAST is a **B**asic **L**ocal **A**lignment **S**earch **T**ool.
- Used for comparing primary biological information viz,
  - Amino acid sequences of protein
  - Nucleotide of DNA or RNA sequences
- BLASTn and BLASTp is particularly used for comparing nucleotide sequence and protein sequence respectively

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Job title: ref|NP\_001276427| (867 letters)

RID [VNKBX2DD015](#) (Expires on 10-09 11:22 am)

Query ID [NP\\_001276427.1](#)  
 Description pikachurin isoform 3 [Mus musculus]  
 Molecule type amino acid  
 Query Length 867

Database Name nr  
 Description All non-redundant GenBank CDS translations+PDB+SwissProt+PIR+PRF excluding environmental samples from WGS projects  
 Program BLASTP 2.8.1+ [Citation](#)

Other reports: [Search Summary](#) [Taxonomy reports](#) [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

**New** Analyze your query with [SmartBLAST](#)

**Graphic Summary**

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Putative conserved domains have been detected, click on the image below for detailed results.

Query seq. 1 125 250 375 500 625 750 867

Cytokine receptor motif

Specific hits FN3 FN3 super LanG LanG superfamily EGF\_CA EGF EGF superfamily EGF\_C Laminin\_6\_2

Superfamilies FN3 super LanG superfamily EGF\_CA EGF superfamily EGF\_C LanG superfamily

Distribution of the top 100 Blast Hits on 100 subject sequences ⓘ

Mouse over to see the title, click to show alignments

Color key for alignment scores

<40 40-50 50-80 80-200 >=200

[Questions/comments](#)

Result of *Paralichthys olivaceus* in BLASTn

Sequences producing significant alignments:

Select: [All](#) [None](#) Selected: 0

[Alignments](#) [Download](#) [GenBank](#) [Graphics](#) [Distance tree of results](#)

|                          | Description                                                                                                            | Max score | Total score | Query cover | E value | Ident | Accession                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------------|---------|-------|--------------------------------|
| <input type="checkbox"/> | <a href="#">Paralichthys olivaceus mRNA for keratin, partial cds</a>                                                   | 662       | 662         | 100%        | 0.0     | 100%  | <a href="#">AB049616.1</a>     |
| <input type="checkbox"/> | <a href="#">PREDICTED: Paralichthys olivaceus keratin, type I cytoskeletal 13-like (LOC109627494), mRNA</a>            | 651       | 651         | 100%        | 0.0     | 99%   | <a href="#">XM_020084024.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Lates calcarifer keratin, type I cytoskeletal 13-like (LOC108875644), mRNA</a>                  | 401       | 401         | 100%        | 7e-108  | 87%   | <a href="#">XM_018664684.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Seriola dumerili keratin, type I cytoskeletal 13-like (LOC111229213), mRNA</a>                  | 385       | 385         | 100%        | 7e-103  | 86%   | <a href="#">XM_022755440.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Mastacembelus armatus keratin, type I cytoskeletal 50 kDa-like (LOC113128586), mRNA</a>         | 318       | 318         | 100%        | 7e-83   | 83%   | <a href="#">XM_026304028.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Mastacembelus armatus keratin, type I cytoskeletal 13-like (LOC113128588), mRNA</a>             | 307       | 307         | 100%        | 2e-79   | 82%   | <a href="#">XM_026304029.1</a> |
| <input type="checkbox"/> | <a href="#">Gasterosteus aculeatus clone CFW18-C12 mRNA sequence</a>                                                   | 296       | 296         | 100%        | 3e-76   | 82%   | <a href="#">BT026740.1</a>     |
| <input type="checkbox"/> | <a href="#">PREDICTED: Stegastes partitus keratin, type I cytoskeletal 13-like (LOC103362534), mRNA</a>                | 285       | 285         | 100%        | 7e-73   | 81%   | <a href="#">XM_008288915.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Amphiprion ocellaris keratin, type I cytoskeletal 13-like (LOC111566298), mRNA</a>              | 274       | 274         | 100%        | 2e-69   | 81%   | <a href="#">XM_023266822.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Austrofundulus limnaeus keratin, type I cytoskeletal 13-like (LOC106518778), mRNA</a>           | 202       | 202         | 100%        | 8e-48   | 77%   | <a href="#">XM_014010178.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Kryptolebias marmoratus keratin, type I cytoskeletal 13-like (LOC108236409), mRNA</a>           | 185       | 185         | 100%        | 8e-43   | 76%   | <a href="#">XM_017417037.2</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Hipposideros armiger keratin, type I cytoskeletal 14 (LOC109395737), mRNA</a>                   | 141       | 141         | 68%         | 2e-29   | 78%   | <a href="#">XM_019667004.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Octodon degus keratin, type I cytoskeletal 14 (LOC101569745), mRNA</a>                          | 135       | 135         | 68%         | 8e-28   | 77%   | <a href="#">XM_004633755.2</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Microcebus murinus keratin, type I cytoskeletal 14 (LOC105872035), mRNA</a>                     | 135       | 135         | 68%         | 8e-28   | 78%   | <a href="#">XM_012765776.2</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Acinonyx jubatus keratin, type I cytoskeletal 17-like (LOC106972591), mRNA</a>                  | 135       | 135         | 68%         | 8e-28   | 77%   | <a href="#">XM_015069589.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Jaculus jaculus keratin, type I cytoskeletal 14 (LOC101596804), transcript variant X2, mRNA</a> | 135       | 135         | 68%         | 8e-28   | 77%   | <a href="#">XM_004655400.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Jaculus jaculus keratin, type I cytoskeletal 14 (LOC101596804), transcript variant X1, mRNA</a> | 135       | 135         | 68%         | 8e-28   | 77%   | <a href="#">XM_004655399.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Ceratotherium simum simum keratin, type I cytoskeletal 14 (LOC101406817), mRNA</a>              | 135       | 135         | 68%         | 8e-28   | 77%   | <a href="#">XM_004434491.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Mus pahari keratin, type I cytoskeletal 14 (LOC110331109), transcript variant X2, mRNA</a>      | 130       | 130         | 68%         | 4e-26   | 77%   | <a href="#">XM_021211493.1</a> |
| <input type="checkbox"/> | <a href="#">PREDICTED: Mus pahari keratin, type I cytoskeletal 14 (LOC110331109), transcript variant X1, mRNA</a>      | 130       | 130         | 68%         | 4e-26   |       |                                |

[Questions/comm](#)

Sequence producing significant alignment

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▼ Next
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 ▲ Descriptions

Paralichthys olivaceus mRNA for keratin, partial cds  
Sequence ID: [AB049616.1](#) Length: 358 Number of Matches: 1

Range 1: 1 to 358
[GenBank](#)
[Graphics](#)

▼ Next Match
 ▲ Previous Match

|           | Score                                                            | Expect | Identities    | Gaps      | Strand    |
|-----------|------------------------------------------------------------------|--------|---------------|-----------|-----------|
|           | 662 bits(358)                                                    | 0.0    | 358/358(100%) | 0/358(0%) | Plus/Plus |
| Query 1   | AAGAACCAGAGAGAACTTGAAGCCTGGCTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG 60  |        |               |           |           |
| Sbjct 1   | AAGAACCAGAGAGAACTTGAAGCCTGGCTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG 60  |        |               |           |           |
| Query 61  | GTGGCAGTCAAGACAGAAAGTTCTCAAACGACCAAGGCAGAAATCTCTGACCTCCGTCGC 120 |        |               |           |           |
| Sbjct 61  | GTGGCAGTCAAGACAGAAAGTTCTCAAACGACCAAGGCAGAAATCTCTGACCTCCGTCGC 120 |        |               |           |           |
| Query 121 | ACAATGCAGAACCTGGAGATCGAGCTGCAGTCTCAACTTAGCATGAAAGCGGCACTGGAA 180 |        |               |           |           |
| Sbjct 121 | ACAATGCAGAACCTGGAGATCGAGCTGCAGTCTCAACTTAGCATGAAAGCGGCACTGGAA 180 |        |               |           |           |
| Query 181 | GGAACGTTGTCAGACACAGAGTGTCTTACTCCATGCAGCTCAAAGCCTCCAGATACAG 240   |        |               |           |           |
| Sbjct 181 | GGAACGTTGTCAGACACAGAGTGTCTTACTCCATGCAGCTCAAAGCCTCCAGATACAG 240   |        |               |           |           |
| Query 241 | GTGACGAGCCTGGAAGAACAGCTGATGCAGCTGCGTCCGACATGGAGCGCCAGAACCAA 300  |        |               |           |           |
| Sbjct 241 | GTGACGAGCCTGGAAGAACAGCTGATGCAGCTGCGTCCGACATGGAGCGCCAGAACCAA 300  |        |               |           |           |
| Query 301 | GAGTACAACATCCTGCTCGACATCAAGACACGGCTGGAGATGGAAATTGCAGAATACA 358   |        |               |           |           |
| Sbjct 301 | GAGTACAACATCCTGCTCGACATCAAGACACGGCTGGAGATGGAAATTGCAGAATACA 358   |        |               |           |           |

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[Graphics](#)

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 ▲ Descriptions

PREDICTED: Paralichthys olivaceus keratin, type I cytoskeletal 13-like (LOC109627494), mRNA  
Sequence ID: [XM\\_020084024.1](#) Length: 1471 Number of Matches: 1

Range 1: 859 to 1216
[GenBank](#)
[Graphics](#)

▼ Next Match
 ▲ Previous Match

|           | Score                                                            | Expect | Identities   | Gaps      | Strand    |
|-----------|------------------------------------------------------------------|--------|--------------|-----------|-----------|
|           | 651 bits(352)                                                    | 0.0    | 356/358(99%) | 0/358(0%) | Plus/Plus |
| Query 1   | AAGAACCAGAGAGAACTTGAAGCCTGGCTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG 60  |        |              |           |           |
| Sbjct 859 | AAGAACCAGAGAGAACTTGAAGCCTGGTTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG 918 |        |              |           |           |
| Query 61  | GTGGCAGTCAAGACAGAAAGTTCTCAAACGACCAAGGCAGAAATCTCTGACCTCCGTCGC 120 |        |              |           |           |

[Gene](#) - associated gene details  
[GEO Profiles](#) - microarray expression data

[Gene](#) - associated gene details  
[New Genome Data Viewer](#) - aligned genomic context

Questions/comm

Sequence alignment of *Paralichthys olivaceus*

# Applications of pairwise sequence alignment

- ✓ Searching large sequences for matches
- ✓ Characterize newly sequenced genes or gene products
- ✓ Molecular distance of evolution between species

# Lunch Break



# Day 2: Sequence Alignment and Phylogenetic

Afternoon Session

# Outline Day 2

## Afternoon (2-5 pm): Phylogenetics

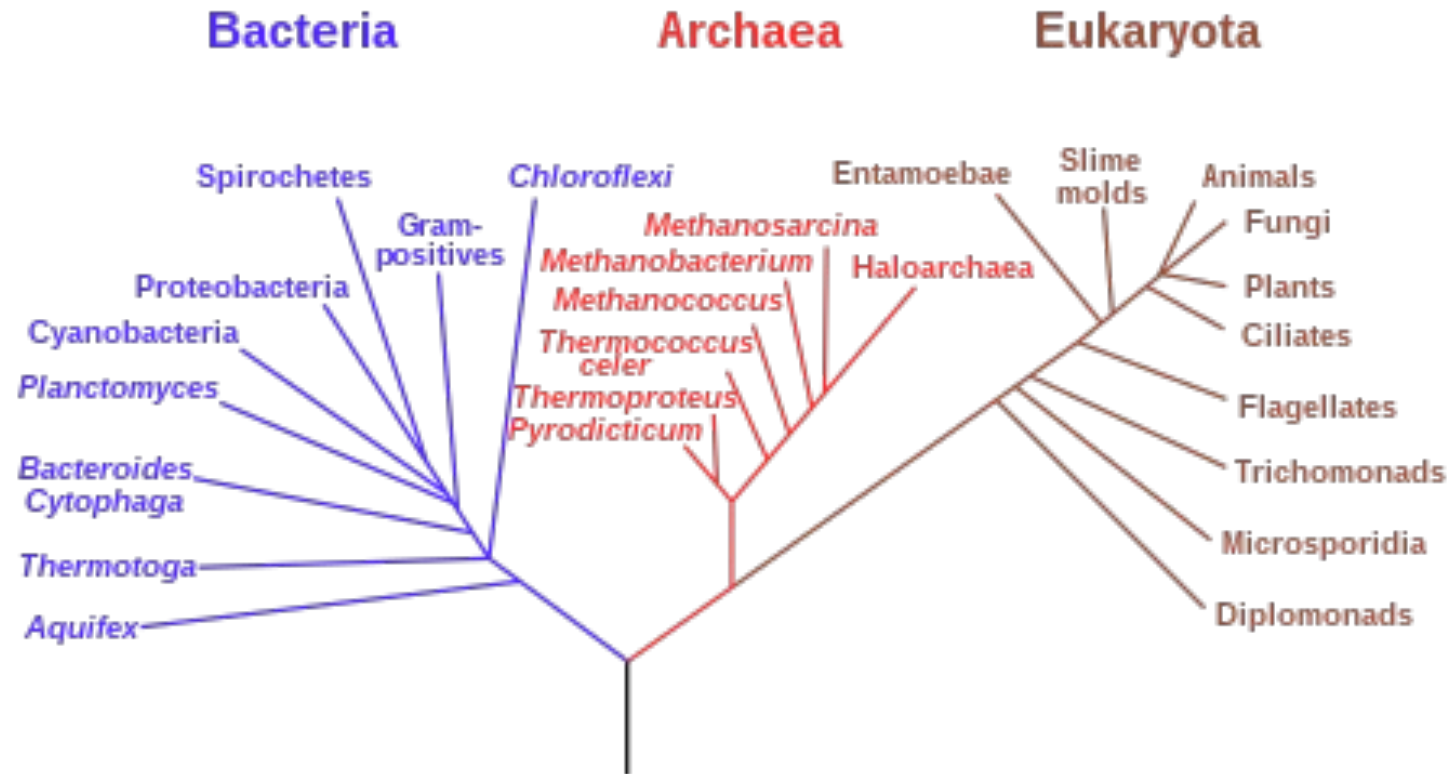
### Phylogenetics:

- Introduction to Phylogenetics tree concept, and significance in evolution biology.

### Practical Session:

- Construct phylogenetics trees using Clustal Omega (web-based) starting from multiple sequence alignments.
- Code with python

# Phylogenetic tree



- A **phylogenetic tree** is a diagram that represents evolutionary relationships among organisms based on the similarities and differences in their genetic and evolutionary characteristics
- The pattern of branching in a phylogenetic tree reflects how species or other groups evolved from a series of common ancestors.
- The phylogenetic tree is also called the “Tree of Life” or “Dendrogram”

# History

- Early representations of "branching" phylogenetic trees include a "paleontological chart" showing the geological relationships among plants and animals in the book *Elementary Geology*, by Edward Hitchcock in 1840.
- Charles Darwin in 1859 also produced one of the first illustrations and crucially popularized the notion of an evolutionary "tree" in his seminal book *The Origin of Species*.



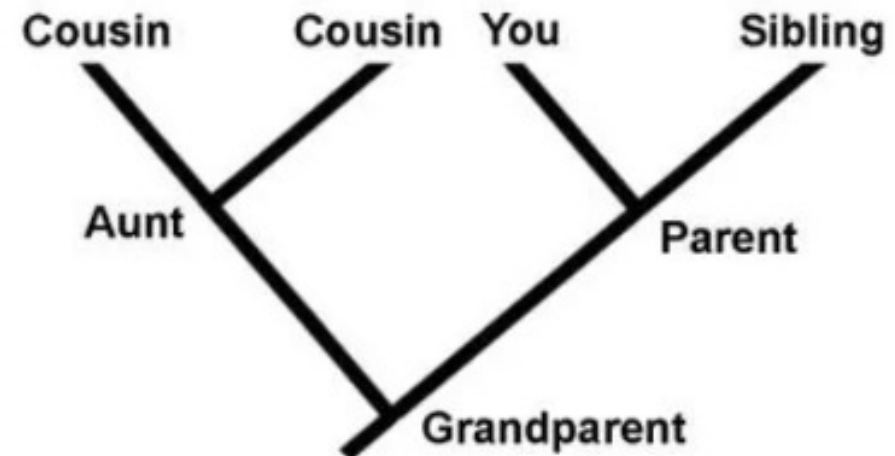
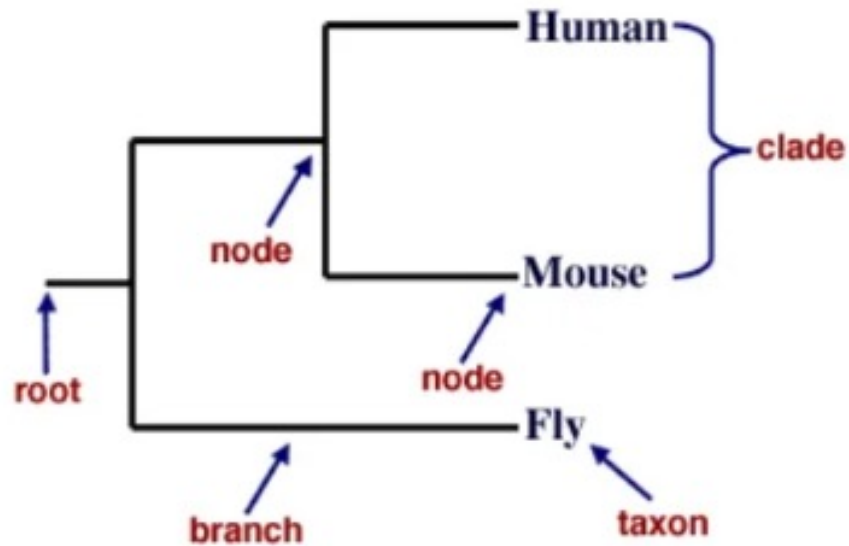
# Importance of Phylogenetic Tree

- It is the fundamental tool to derive their most-useful evidence from the fields of anatomy, embryology, palaeontology and molecular genetics. Other significances of the phylogenetic tree are:
- Used in the search for a new species.
- Used to study evolutionary histories.
- To study how the species were spread geographically.
- To study the common ancestors of extant and extinct species.
- It is used to identify the most recent common ancestors and to recognize how closely related species are.
- To relate the milestones of the evolution of major life forms to the tree of life.
- To represent evolutionary relationships between organisms that are believed to have some common ancestry.
- With the help of the phylogenetic tree, the infectious microbes can be traced along with their evolutionary histories.

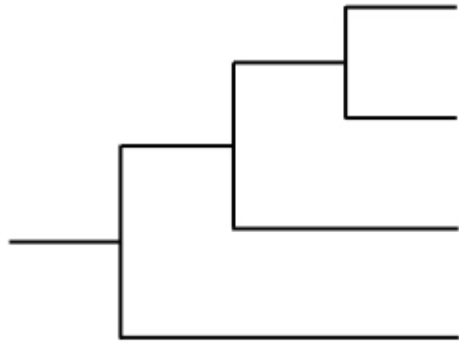
| Term             | Explanation                                                                |
|------------------|----------------------------------------------------------------------------|
| Phylogeny        | A method to construct a phylogenetic tree                                  |
| Common ancestors | A group of organisms sharing the common feature with the decedent.         |
| Taxon            | An organism of the entire taxa                                             |
| Taxa             | A group of organisms from a species or from a different species.           |
| Node             | Nodes represent the common ancestors of Different taxons.                  |
| Sister group     | Two or more taxon share the same node.                                     |
| Outgroup         | The taxon is outside the interest group or other than the common ancestor. |
| Clade            | A clade is a group of all organisms from a common ancestor.                |

# What does this tree looks like?

➤ What do the lines represent?

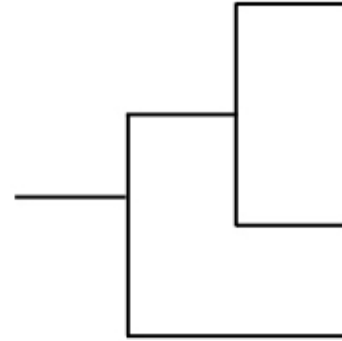






**A**

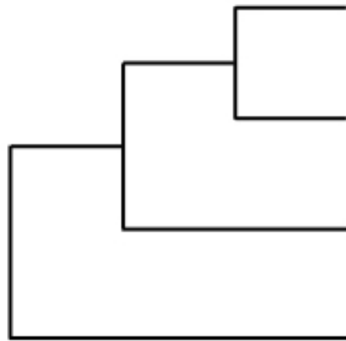
Rooted tree



**C**

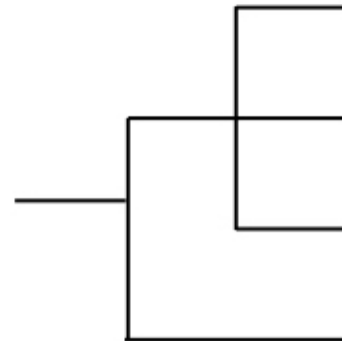
Bifurcating tree

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**B**

Unrooted tree

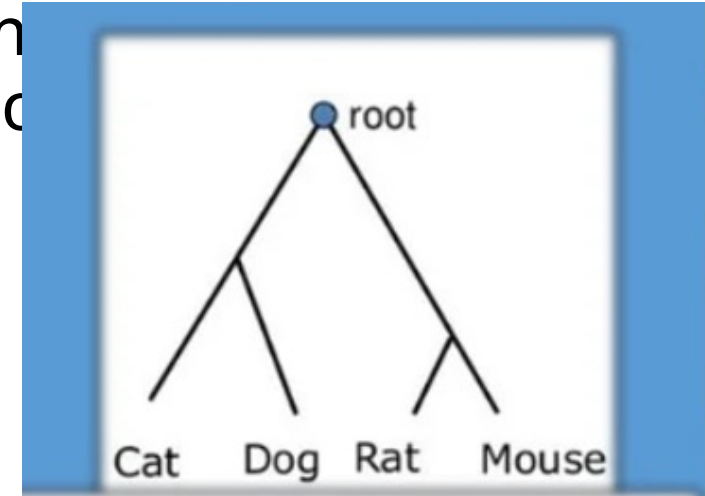


**D**

Multifurcating tree

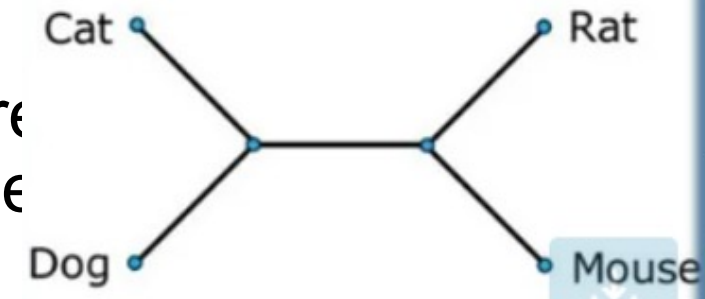
## Rooted tree:

- The rooted tree is described as a phylogenetic tree sharing the common ancestor on the node. Therefore the classification ends at one point usually on the node which is the common ancestor of all the branches of the tree.



## Unrooted tree:

- Contrary to the rooted tree, the non-rooted tree doesn't have a common ancestor. The unrooted phylogenetic tree is always prepared from the rooted tree by excluding the common ancestor or the node of the tree.



## Bifurcating tree

- The phylogenetic tree only has two branches or we can say leaves are known as bifurcating trees. It is also classified in rooted bifurcation trees and unrooted bifurcating trees.

## Multifurcating tree:

- The multifurcating tree is described as having multiple branches on a single node. Again, it is classified into a rooted multifurcating tree and an unrooted multifurcating tree.

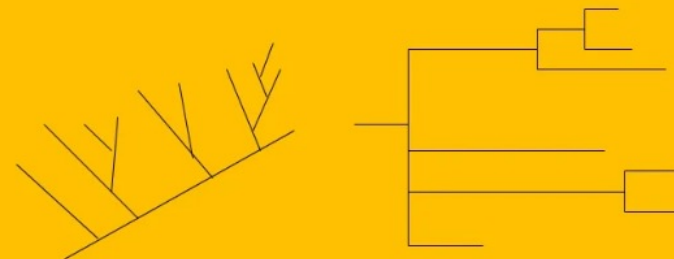
### The Bifurcating Tree

- A tree that bifurcates has a maximum of 2 descendants arising from each of the interior nodes.



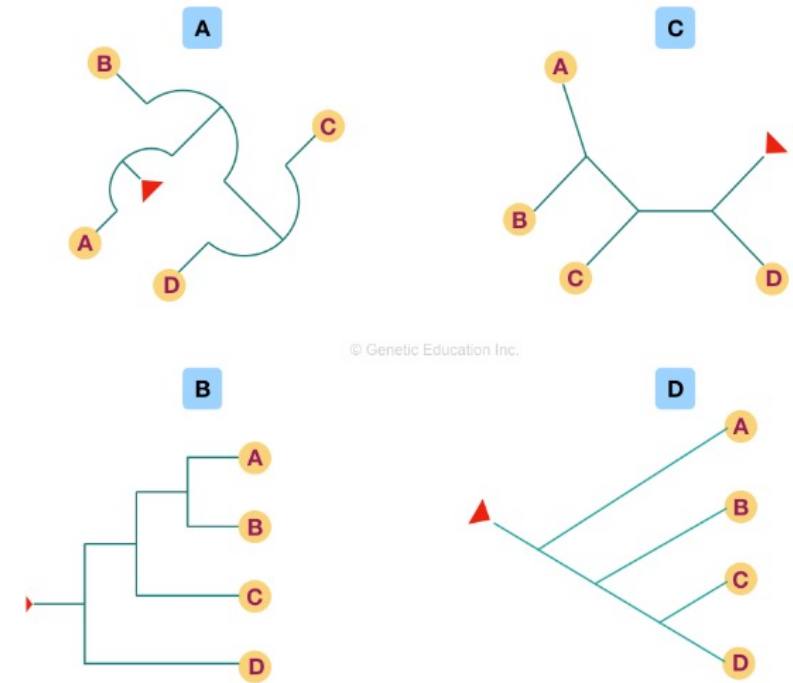
### The Multi-furcating Tree

- A tree that multi-furcates has multiple descendants arising from each of the interior nodes.



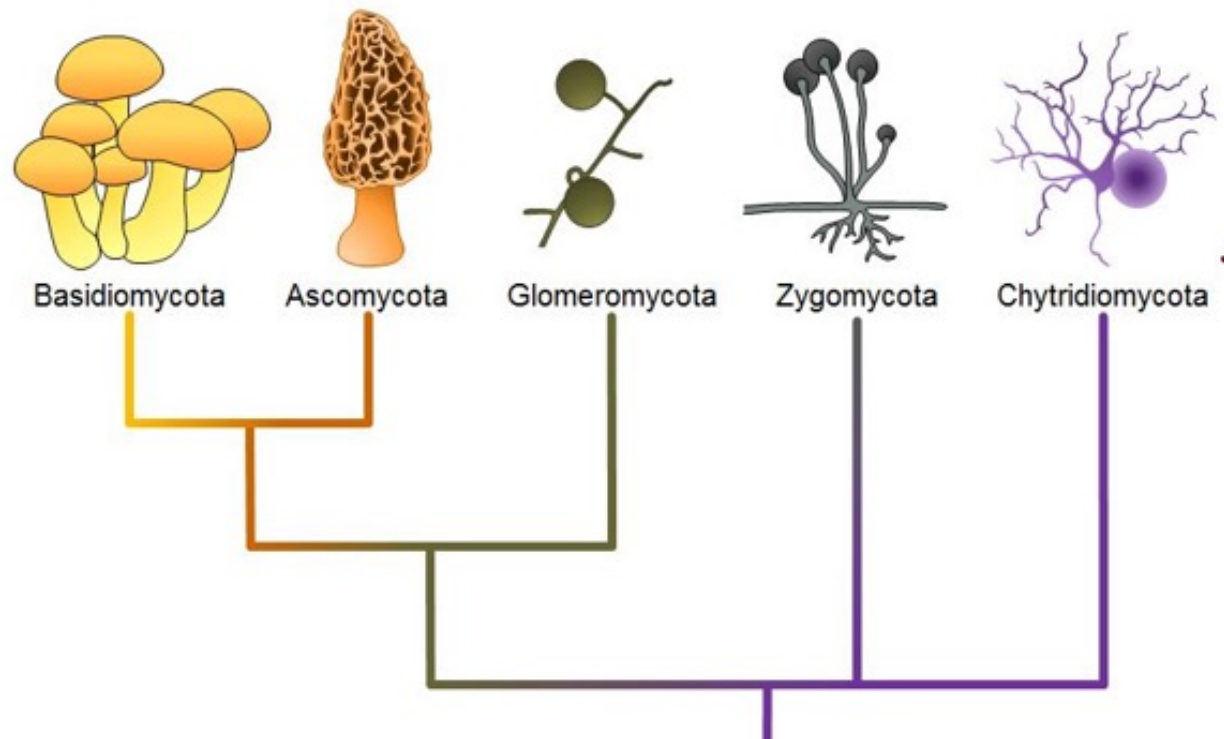
In phylogeny, the node is also known as a “clade” as well. Though there are so many different variations of the phylogenetic tree, every method of making a tree depicts the same type of information. Take a look at various trees shown in the figure

Keep in mind that whatever the shape, topology or structure of the tree is, it must have a common node if rooted and branched. To read a tree, start with the tip of the branches and see where the branch ends (the node), based on that information you can depict or conclude which organism is nearer or closer and which are distantly related.



*The figure represents different forms of a single type of phylogenetic tree.*

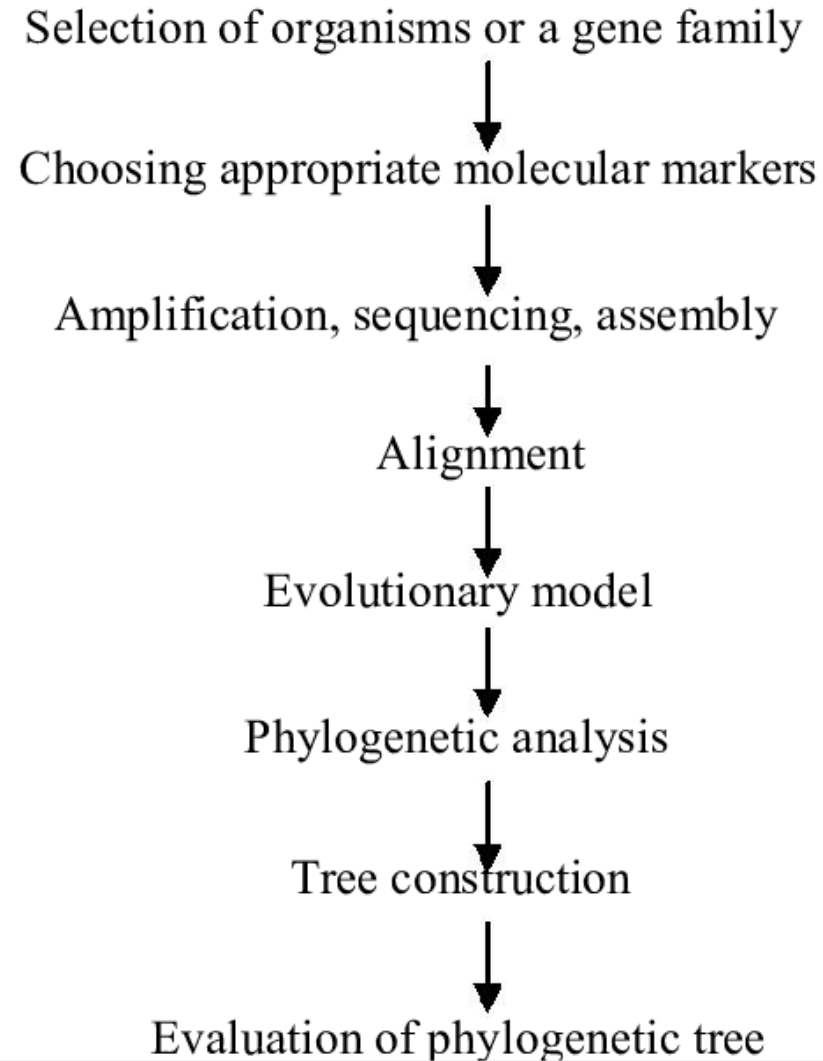
# Phylogeny in fungus



# Applications of a phylogenetic tree:

- The phylogenetic tree is constructed to make an evolutionary link between various organisms. By doing so, we can get an idea about how and from whom different organisms are evolved.
- Also, it helps to classify organisms and species in different taxa and groups based on their DNA sequence and phenotypic similarities and differences.
- In addition to this, it is useful to study the force of evolution and characteristics of different organisms.
- it is applicable to study the events occurring during the course of evolution and to classify species based on the divergence of structure and function.

## Steps in Phylogenetic Analysis



# Software

- Some programs for phylogenetic analysis
- A multiple alignment program:
  - Clustal, T-Coffee, MAFFT, Muscle...
- A phylogenetic program:
  - Phylip, PAUP\*, MacClade, BioEdit...
- Visualizing the tree:
  - TreeView, Njplot
- **<https://evolution.genetics.washington.edu/phylip/software.html>**



# Selecting sequence

- The rate of mutation is assumed to be the same in both coding and non-coding region
- However there is a difference in substitution rate
- Non-coding DNA region have more substitution than coding regions.
- Protein are much more conserved since they need to conserve their function
- It is better to use sequence that mutate slowly (protein) than DNA. If the gene are very small or they mutate slowly, then it can use for building tree.

# Building Phylogenetic Trees

- The most popular and frequently used methods of tree building can be classified into two major categories
- Phenetic methods based on **distances**
- Cladistic methods based on **characters**
- **Distance matrix methods**
  - UPGMA (Unweighted pair group methods with arithmetic mean)
  - Fitech-Margoliash
  - Neighbour joining (NJ)
- **Character based methods**
  - Maximum parsimony (MP)
  - Maximum likelihood (ML)

# Distance based methods

- Tree are calculated by similarities of sequences and are based on distance
- Some sequences more similar than others
- Closely related sequences should be close in the tree
- Only use the distances between sequences
- All methods start with a *distance matrix*

# UPGMA Vs Fitch Margoliash Method

## UPGMA Methods

Unweighted Pair Group Method with Arithmetic Mean

Unweighted: The distances are used as they are

Pair: Find the two closest elements

Group: Put them together in a new group

Arithmetic Mean: Gives distances from the new group

## Fitch-Margoliash Methods

More complicated than UPGMA

Does not assume a molecular clock

Produces an unrooted scaled tree

# Neighbour joining

- This method tries to correct the UPGMA method for its assumption that the rate of evolution is the same in all taxa.
- But it assumes an additive tree
  - Distance between two leaves is the sum of the edges
- Find the *closest pair* that is *most apart* from the rest of the tree
- Connect pair and update distances
  - A little advanced: Take the overall distance to the rest of the tree into account
  - Corrects for varying mutation
- Fast and can give good results

# Character methods

- Tree are calculated by considering the various possible pathways of evolution.
- This methods uses each alignment positions as evolutionary information to build a tree.
- All information at hand
- More advanced, slower, but also more accurate
- Maximum Parsimony (MP)
  - Occam's razor: Simplest explanation
- Maximum Likelihood (ML)
  - Advanced statistical method
  - Most probable tree given the data and the model

# Maximum parsimony (MP)

- For each position in the alignment all possible trees are evaluated and are given a score based on the number of evolutionary changes.
- More time consuming
- Used for closely related sequences
- The most parsimonious tree is the one with the fewest evolutionary changes
- MP methods are available for DNA in Programs **paup**, **molphy**, **phylo\_win**

# Maximum Likelihood

- This method also uses each position in an alignment, evaluate all possible trees and calculates the likelihood for each tree.
- The tree with the maximum likelihood is the most probable tree.
- Slowest method but gives the best result
- Used for any set of sequence.
- Maximum likelihood methods can be found in **phylip, paup or puzzle**